

5	(a)	Twin core cable, consisting of a 'live' wire and a 'neutral' wire, is commonly used for wiring applications. Current is supplied via the 'live' wire from an electrical power source to the appliance. Current then flows via the 'neutral' wire back to the electrical power source. The twin core cable shown in Fig 5.1 consists of two thin wires whose centres are 5.0 mm apart. Each wire carries a current of 0.25 A. Permeability of the insulating material, $\mu = 1.3 \times 10^{-6} \text{ H m}^{-1}$. Fig 5.1	
	(i)	State whether the force between the wires is attractive or repulsive.	
			[1]
		Solution: Repulsion	1

(ii) Calculate the magnitude of the force per unit length on each wire.

force per unit length =N m⁻¹ [3]

Solution:

The magnetic flux density at one wire due to the other wire is,

$$B = \frac{\mu I}{2\pi d}$$

$$= \frac{1.3 \times 10^{-6} \times 0.25}{2\pi \times 5 \times 10^{-3}} \text{ T}$$

Force on each wire = BIL

Force per unit length = BI

$$= \frac{1.3 \times 10^{-6} \times 0.25}{2\pi \times 5 \times 10^{-3}} \times 0.25$$

$$= 2.5863 \times 10^{-6} = 2.6 \times 10^{-6} \text{ N m}^{-1} \text{ (2 s.f.)}$$

- (b) A railgun is a device typically designed as a weapon that uses electromagnetic force to launch high speed projectiles. A simplified model of a railgun is shown in Fig. 5.2. Fig. 5.2 shows the top view of a pair of smooth horizontal metal rails connected across a 2.0 V battery of negligible internal resistance. The metal rails have negligible resistance. A copper rod XY of resistance 3.04 mΩ is resting on the rails. The rails and the rod lie in a horizontal plane and a uniform vertical magnetic field of flux density 24 mT is applied out of the page.

When the switch is closed, a magnetic force acts on XY which causes it to start moving towards the right. At time t s after the switch is closed, XY has a velocity v m s⁻¹.

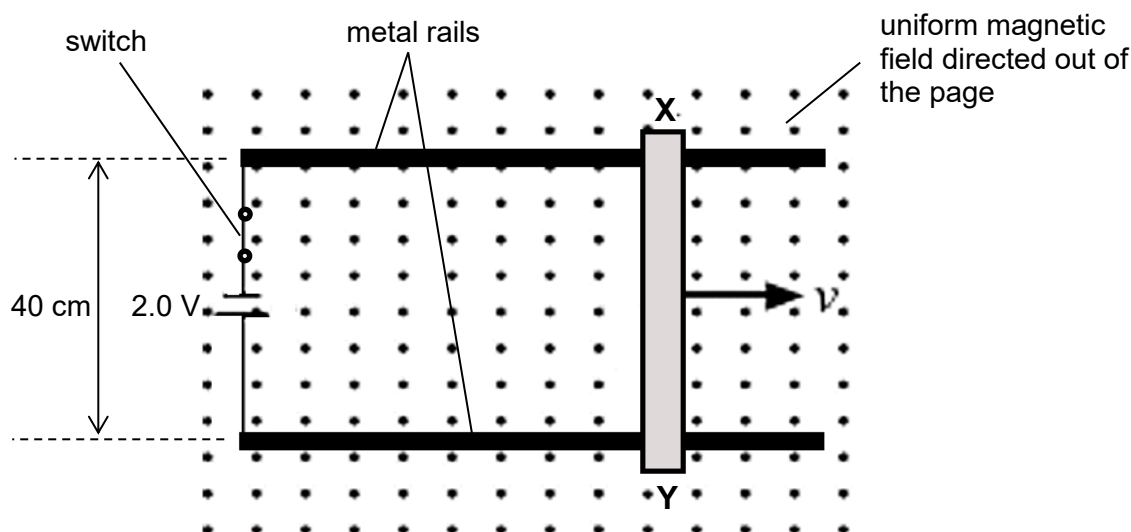


Fig. 5.2

		(i)	As the rod moves to the right, there is an induced e.m.f across rod XY. Use Lenz's law to explain why the induced electric potential at X is lower compared to Y.	
				
				[2]
			Solution: As rod XY moves, it continuously sweeps an area perpendicular to the uniform magnetic field. This causes the rod to experience an increasing magnetic flux linkage through the area swept. [OR As the rod moves, the magnetic flux linkage through the area enclosed by the closed circuit, consisting of the battery and rod increases.) By Lenz's law, the induced e.m.f. in XY will attempt to drive a current in XY that will flow in a direction that induces a magnetic field directed opposite (i.e. into the page) to the original uniform magnetic field to reduce this increase in magnetic flux linkage. This causes X to be at a lower potential than Y.	1 1
		(ii)	Show that the current, in amperes, in the closed circuit at time t is given by $(660 - 3.2v)$.	
				[2]
			Solution: Net e.m.f = $2.0 - \text{back induced e.m.f}$ $= 2.0 - BLv$ $= 2.0 - (24 \times 10^{-3})(40 \times 10^{-2})v$ Current in closed circuit = Net e.m.f / circuit resistance $= [2.0 - (24 \times 10^{-3})(40 \times 10^{-2})v] / (3.04 \times 10^{-3})$ $= 657.89 - 3.1579 v$ $= 660 - 3.2v$	1 1

		(iii)	Show that after some time rod XY will slide with constant velocity. Determine this velocity. Explain your working.	
			velocity =m s ⁻¹	[4]
			Suggested solution 1: The current in the closed circuit is equal to $(660 - 3.2v)$, as the rod speeds up, v increases and therefore current decreases. Eventually, when the current reduces to zero. Thus the net force provided by the magnetic force (F) acting on the rod XY also reduces to zero. Acceleration ceases, and the rod XY continues moving with constant velocity. $I_{\text{net}} = 660 - 3.2v = 0$ $v = 210 \text{ m s}^{-1}$ Suggested solution 2: Initially, a net current flows anticlockwise, given by $I_{\text{net}} = \frac{E - BLv}{R} = 660 - 3.2v \text{ ---- (1)}$ Where E is the battery's e.m.f., B is the magnetic flux density, L is length of the rod XY, and R is the circuit resistance. At the instant when speed of rod XY is v, Magnetic force on rod XY acting to the right, $F = B I_{\text{net}} L$ $= \frac{E - BLv}{R} BL$ As the rod XY speeds up, v increases, causing induced current (I_{induced}) to increase, while the current driven by the battery (I_{battery}) remains constant. Thus the net current (I_{net}) flowing anticlockwise through the closed circuit decreases with time. [See equation (1).] Eventually, when the induced e.m.f. (BLv) increases to a value equal to the e.m.f. of the battery (E), I_{net} reduces to zero. Thus the net force provided by the magnetic force (F) acting on the rod XY also reduces to zero. Acceleration ceases, and the rod XY continues moving with constant velocity. Therefore, at maximum speed, $E = BLv$ $\therefore v = \frac{E}{BL} = \frac{2}{(24 \times 10^{-3})(40 \times 10^{-2})} = 208.33 = 210 \text{ m s}^{-1}$	1 1 1 1

6	A photocell is connected in a series circuit with a variable d.c. power supply and a sensitive
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